DHRUV SHARMA

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EDUCATION

- BTech – Computer Science Engineering, Manipal Institute of Technology, Udupi, Karnataka (Batch 2024 - 2028)
  CGPA: 8.68/10 (till 4th Semester) | Year: December 2025

SKILLS

- **Languages:** C, Python (NumPy, Pandas, Basic ML workflows), Java, SQL (MySQL)
- **Developer Tools:** Git, Github
- **Machine Learning and Data Science:** Data Cleaning and Exploratory Data Analysis (EDA), Machine Learning Fundamentals (Classification, Regression, Transformers, XAI Applications), Logistic Regression, Neural Network, Model Preprocessing and Evaluation
- **Cloud:** AWS (EC2, RDS)

EXPERIENCE

Undergraduate Researcher *Aug 2025 – May 2026*

At Manipal Institute of Technology, Udupi, Karnataka Under the guidance of Assistant Professor Dr. Krishnaraj Chadaga

- Conducting research on machine learning and explainable AI methods for clinical decision support in seizure classification.
- Designing and implementing data preprocessing pipelines, model training workflows, and evaluation frameworks for healthcare datasets.
- Applying machine learning, deep learning, and explainable AI techniques to analyze clinical data and interpret predictive models.
- Collaborating on research documentation, experimental analysis, and submitted results for research publication.

PROJECTS

Seizure Classification | NumPy, Pandas, Scikit-learn, Transformers, XAI

- Developed an ML-based clinical decision-support framework to distinguish epilepsy-positive cases from psychogenic nonepileptic seizures (PNES) using clinical EMU data from 271 patients (24 engineered features).
- Evaluated 8+ ML/DL models, including Random Forest, XGBoost, Logistic Regression, Neural Networks, and Transformer architectures.
- Applied Explainable AI techniques (SHAP, LIME, DiCE, PDP, ICE, Anchors) to generate global and local explanations for clinical interpretability.
- Achieved ROC-AUC ≈ 0.94 (Random Forest) and ≈0.92 (XGBoost) while improving minority-class detection through SMOTE and hyperparameter optimization

Support & Contrast Learning Platform | HTML, CSS, JavaScript, Vercel

- Designed and developed an interactive learning platform for GRE verbal preparation featuring flashcards, reverse flashcards, fill-in-the-blank quizzes, and categorized reference modes.
- Organized hundreds of logical transition words into semantic categories including support, contrast, negation, and emphasis to improve sentence reasoning skills.
- Implemented progress tracking, keyboard shortcuts, and multiple study modes to create a distraction-free learning experience.
- Deployed the platform publicly and grew it to nearly 1,000 users, demonstrating real-world adoption and product usability.

Student Project Inventory Management System | Full-Stack Development, SQL

- Built a centralized inventory management platform supporting 10+ student engineering projects, streamlining the tracking and allocation of 300+ electronic components.
- Designed and implemented a relational database with 10+ tables for stock management, issue/return workflows, and project-wise inventory records.
- Developed a responsive interface that reduced duplicate requests and manual tracking for teams coordinating primarily through WhatsApp.
- Collaborated in a team environment to integrate backend logic with the user interface, enabling real-time inventory updates.

ACHIEVEMENTS

Achievers Scholarship — Manipal Institute of Technology 2025

- Awarded for academic excellence: ranked **8th out of 233 students** (Top 10%) in the branch during first year

Awarded National Talent Search Scholarship 2020

- Conducted by NCERT to identify and nurture talented students
- Acceptance rate of < 0.1% out of ~ 1,000,000 applicants

CERTIFICATION

- Neural Networks and Deep Learning (Coursera, Issued in June 2025)